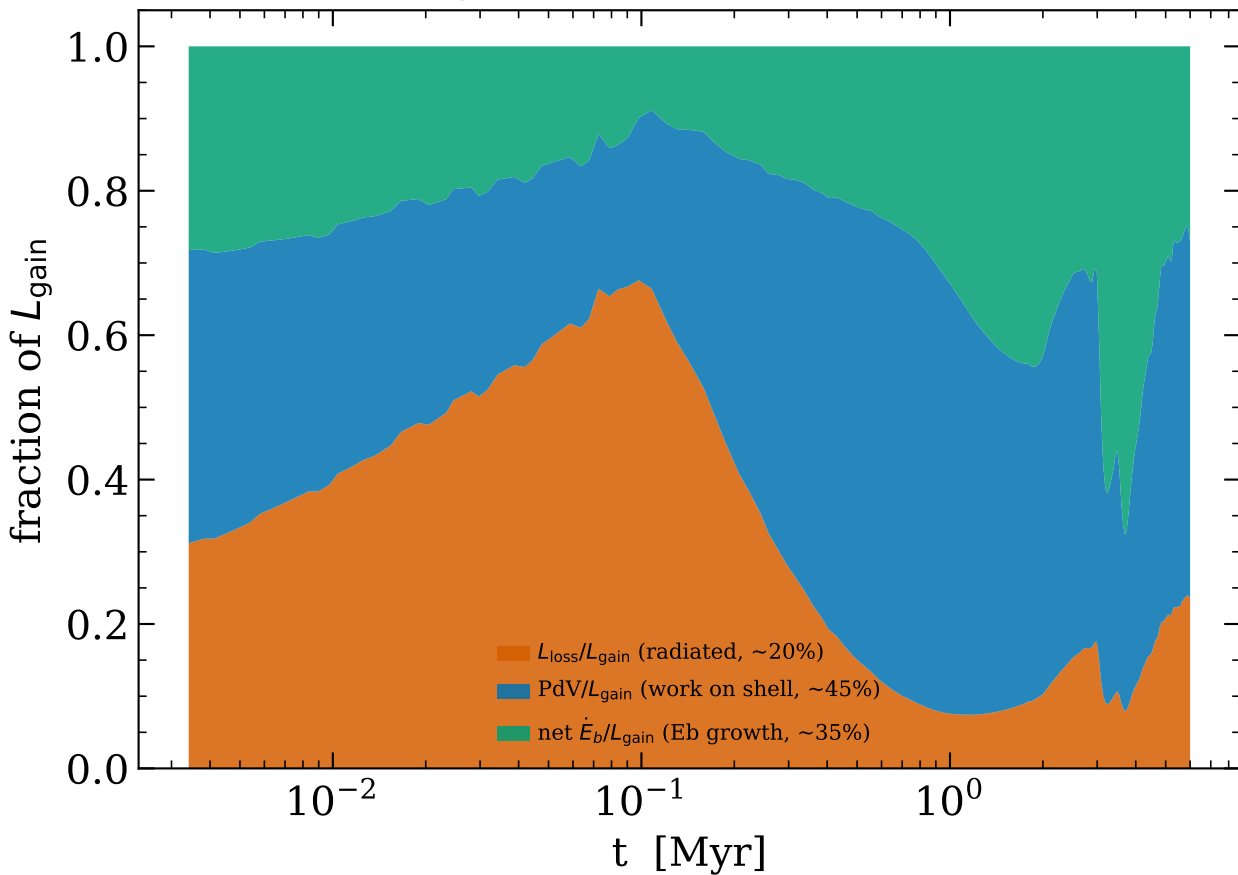


PdV is the energy-DRIVING work ($\sim 45\%$ of input), not a loss — putting it in the trigger asks 'has Eb peaked?', not 'is cooling winning?'

Where the input goes (simple_cluster):
nearly half is expansion WORK, not radiated



Adding PdV pulls $\sim 0.5 \rightarrow \sim 0.05-0.15$ (nearly fires),
but it measures the Eb-peak, not cooling — still never < 0.05 (5/6)

